

Welcome to MedTech Prototyping Skills!

Prototyping Skills (BME290L)

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Outline

Get to Know Each Other

Who am I?

Who are you?

Prototyping Skills

Functional Decomposition

CAD

EDA

Syllabus Review

Resumes / Portfolios

Resumes

Online Portfolios

Lab

Who am I?

Long Duke track record...

- ▶ Duke BME/ECE (1996–2000)
- ▶ Duke BME Grad (2005)
- ▶ Duke Med (2007)

Who am I?

Teaching

- ▶ BME153L/154L (Introduction to Circuit Analysis)
- ▶ BME354L (Introduction to Medical Instrumentation)
- ▶ BME547 (Medical Software Design)
- ▶ BME473L/474L (Medical Device Design)
- ▶ Founded BME Design Fellows program
- ▶ BME290L (MedTech Prototyping Skills)

Who am I?

Research

- ▶ Acoustic radiation force
- ▶ Ultrasonic elasticity imaging (non-invasive liver fibrosis, prostate cancer, MSK)
- ▶ Finite element method modeling
- ▶ Deep learning (signal and image processing) (POCUS)
- ▶ Medical devices for LMICs
- ▶ `https://palmeri.io`

Who am I?

Fun



Who are you?

Who are you?

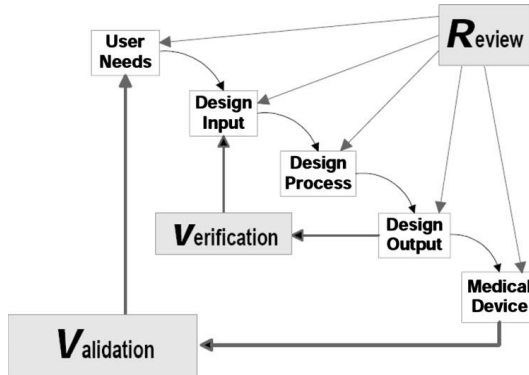
- ▶ Name
- ▶ Hometown
- ▶ What do you aspire to do after Duke?
- ▶ Passion / memorable fact

Why skills?

- ▶ Do not want design ideas to be limited by what you “think” you can actually make!
- ▶ As the engineer, you bring both solution creativity and execution to the table.
- ▶ Most of the engineering curriculum has been “book heavy” to date.
 - ▶ Functional Decomposition
 - ▶ CAD/EDA (simulation, PCB layout)
 - ▶ Fabrication & Integration (3D printing, soldering, fastening, mounting)
 - ▶ Verification & Validation: testing to specifications
 - ▶ MCU Firmware Development (C, `git`)

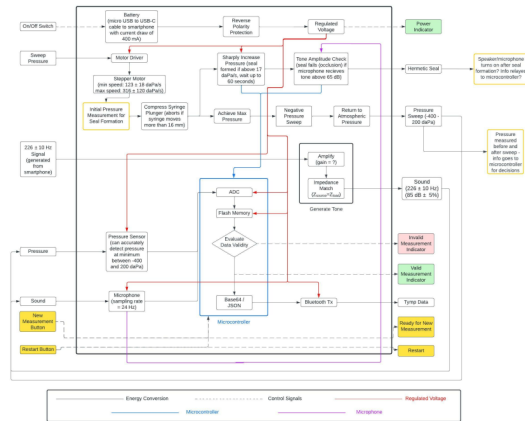
Functional Decomposition

FDA V&V Waterfall



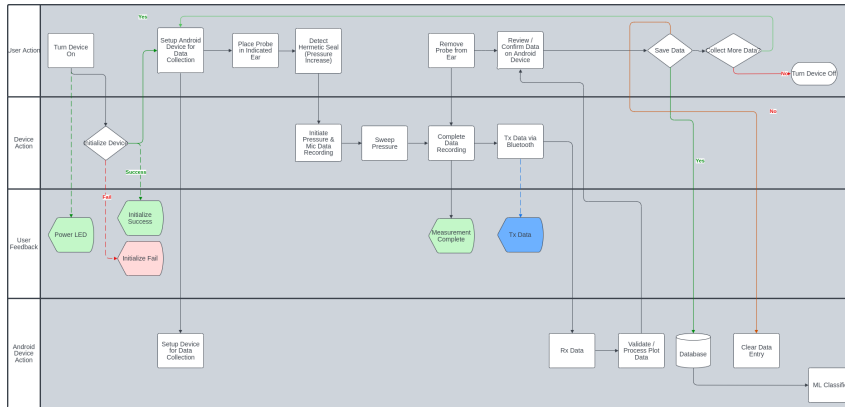
Functional Decomposition

Functional Decomposition



Functional Decomposition

User Action Flowchart



CAD: Computer Aided Design

Why CAD?

- ▶ Formally capture 2D projection sketches → 3D parts.
- ▶ Dimension consistency checks.
- ▶ Assemble multiple parts to check fit, interfaces and simulate stress/strain/movement.
- ▶ Translate to physical realization (e.g., 3D print, mil).
- ▶ General technical drawings for manufacturing.
- ▶ Capture design history, while facilitating iteration / change.

onshape



- ▶ We will be using onshape, a cloud-based CAD package with a similar workflow to SolidWorks.
- ▶ You will be automatically registered to join Duke's onshape team (<https://duke.onshape.com>).
- ▶ Tutorials are available through <https://learn.onshape.com> (will be explicitly assigned).

mHealth Tympanometer



EDA: Electronics Design Automation

Why EDA?

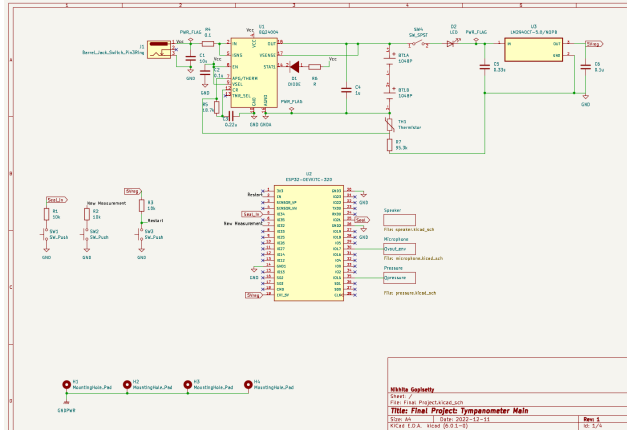
- ▶ Formalize circuit schematics.
- ▶ Validate circuit behavior (design rules and SPICE simulations).
- ▶ Convert circuit to printed circuit board (more space efficient and permanent than a breadboard).
- ▶ Capture design history and facilitate rapid iteration.
- ▶ Generate 3D “parts” to integrate with CAD.

KiCad

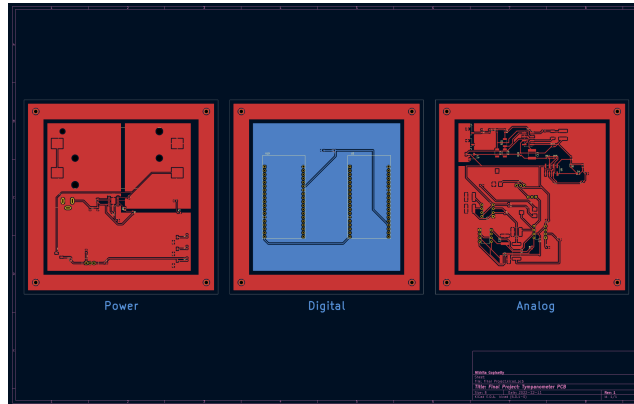


- ▶ <https://kicad.org>
- ▶ A completely open-source alternative to Eagle.
- ▶ Just as capable as Altium Designer (to a point).

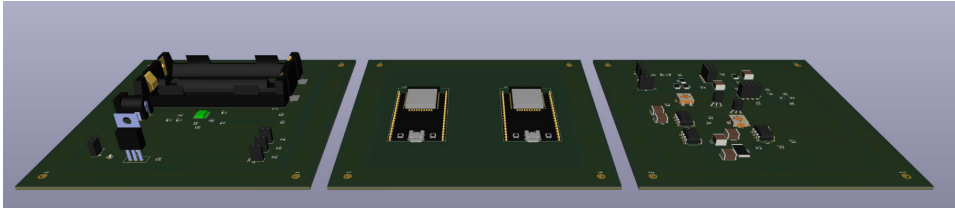
Schematic Capture



PCB Layout



PCB 3D Model



Syllabus

- ▶ “Live” copy maintained on Sakai (will include revision number / compilation date)
- ▶ Many hyperlinks embedded in syllabus.
- ▶ Lectures slides will be posted on Sakai before lecture.
- ▶ Assignments and lab exercises will be highlighted in lecture.
- ▶ Detailed information, including due dates, posted on Gradescope.

Resumes - make them more appealing and memorable

EDUCATION

Duke University, Class of 2021
Bachelor of Science in Biomedical Engineering, Minor in Chemistry

On-Campus Involvement

- E-Team Peer Advisors**, Freshman Mentor, 2018-2020
- Society of Women Engineers**, Active Member, 2019-2020
- Duke BME Design Fellows**, Active Member, 2019-2020

SKILLS

- Programming Languages**: MATLAB, Python, and SQL
- Practical Design Skills**: Agile Management Tools, Brainstorming with Mind Mapping, CAD (Fusion360, TinkerCAD), ECAD (Circuit Simulation, PCB Layout)
- Adult and Pediatric CPR, AED, & First Aid Certified**, 2020
- Healthcare Innovation & Entrepreneurship Honors Certificate**, Coursera, 2018
- Conversational Spanish**

INTERESTS

- Duke Club Basketball Team**, President/Captain, 2017-2020
- Duke Recreation Outdoor Hiking Adventures**, Member, 2018-2020
- Spoon University Duke Chapter**, Food Photographer, 2020

LINKS

- [LinkedIn.com/in/\[redacted\]](#)
- [GitHub.com/\[redacted\]](#)

WORK EXPERIENCE

Digitouch Health, Valhalla, NY
Product Research & Development Intern • Summer 2020

- Improved MATLAB software of a blood pressure device by organizing datasets, filtering raw and pressure signals, and detecting and removing pressure artifacts.
- Processed raw clinical trial device data, blood pressure references, finger images, and subject demographics using improved BP device algorithm.
- Investigated the effects of hand position, finger misalignment, finger movement, LED Power, skin height, and distance between the emitter and photodiode on blood pressure measurement accuracy and precision through multiple experiments.

Antitake Test Prep, Online via Skype
General Manager & Tutor • Fall 2018 - Present

- Tutored five students in SAT/ACT, Algebra I, & Chemistry (>20% score increase).
- Trained two new tutors and developed and lowered them via weekly phone calls.
- Created all company documentation, such as non-competitive forms, rule books, training manuals, non-competitive guides, email templates, and teaching presentations.

VA Freedland Lab, Durham VA Hospital, Durham, NC
Data Technician • Fall 2019 - Spring 2020

- Detected over 700 errors in the VA database system using SQL, server.
- Updated instructions for manual data entry checks and increased the Research Management System (RMS) database manual.

NY Institute of Sports Medicine, White Plains, NY
Research Intern & Shadowing • Summer 2019

- Collected our movement data from 12 patients to assess the efficacy of the Cybex technology and to gain insight on knee concerns.
- Shadowed orthopedic surgeons, including meniscus tears and ACL reconstructions.
- Served for a pain management specialist and assisted in image-guided systems.

Well Cornell Medicine, New York, NY
Clinical Research Assistant • Summer 2018

- Completed 7 IRB requests for the Transplantation Oncology Infectious Diseases Department for diagnostic trials such as FOLFIRI and Imatinib in Adenocarcinoma.
- Tracked the spread of HRS by webbing ICU employee gloves and lab coats.

VOLUNTEER WORK

FEMMES Connect, Duke University
Director • Fall 2018 - Spring 2020

- Planned 45 Durham high school students with 40 Duke members.
- Organized 12150 needed events such as visits to the senior center, resume seminars, engineering design events, coffee chats, and coding workshops.

Zeta Tau Alpha, Phi Chapter, Durham, NC
Director of Philanthropy • Fall 2018 - Spring 2020

- Raised over \$2500 for ZTA's Think Pink Breast Cancer Foundation.
- Spoke awareness through 18 color run, coffee chats, sales, a basketball tournament, selling handmade pins, and hosting dinners at local restaurants.

American Red Cross Club, Duke University
Active Member • Fall 2018 - Fall 2020

- Raised over \$1200 for the "You are in Charge" campaign.
- Spoke awareness of measles & rubella in most All North through lab coats, selling light pins, and an informational flyer campaign.

Online Portfolio

- ▶ Including links to online work product is great (e.g., <https://github.com/mlp6>).
- ▶ Showcasing your work is even better (e.g., <https://suyashkumar.com/>).
- ▶ Something to work on as you look to “sell” yourself for applying for internships, etc.

Lab This Week

- ▶ Consider registering for the Friday morning session!
- ▶ Start at 14:00 (no food/drink in POD)
- ▶ Install software on personal laptops
- ▶ Confirm function
 - ▶ Setup git to clone & push
 - ▶ Load KiCad project
 - ▶ Log into Onshape
- ▶ Order MCU